

SAT UZBEKISTAN

Telegram @satuzbhunters

Student-produced response directions

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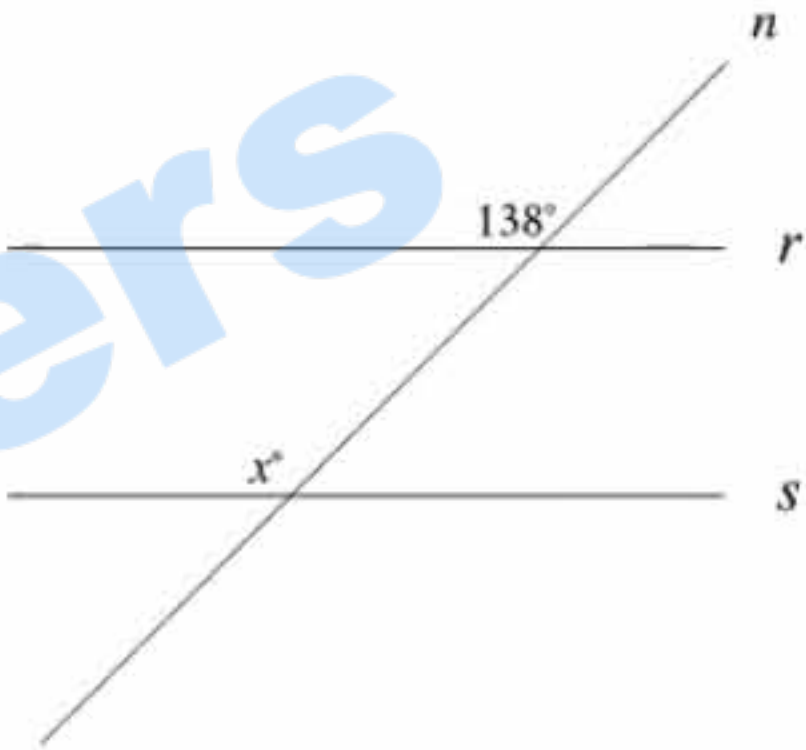
Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 ~ ~ ~ ~	0.66 .66 0.67

1



Mark for Review



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

- A) 138 B) 32 C) 42 D) 52

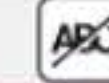
Answer Preview:



2



Mark for Review



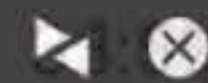
During a portion of a flight, a small airplane's cruising speed varied between 125 miles per hour and 135 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

(A) $s \leq 10$

(B) $s \leq 125$

(C) $s \leq 135$

(D) $125 \leq s \leq 135$



Hide



Calculator



Reference



More

3



Mark for Review



If $5x = 4$, what is the value of $40x$?

(A) 4

(B) 12

(C) 28

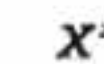
(D) 32



Hide



Calculator



Reference



More

4



Mark for Review



Which expression is equivalent to $(x^3 + 4x^2 - 3x) + 5(x^2 + 8)$?

(A) $x^3 + 9x^2 - 3x + 40$

(B) $x^3 + 4x^2 - 3x + 40$

(C) $x^3 + 8x^2 - 3x + 40$

(D) $x^3 + 5x^2 - 3x + 40$



Hide

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5

Mark for Review

The function f is defined by $f(x) = x^3 + 14$. What is the value of $f(2)$?

A) 20 B) 22 C) 24 D) 26

Answer Preview:



Hide

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6

Mark for Review

The function g is defined by $g(x) = \frac{x}{2}$. For what value of x does $g(x) = 624$?

1248 B) 1296 C) 1312 D) 1316

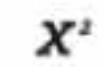
Answer Preview:



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Calculator

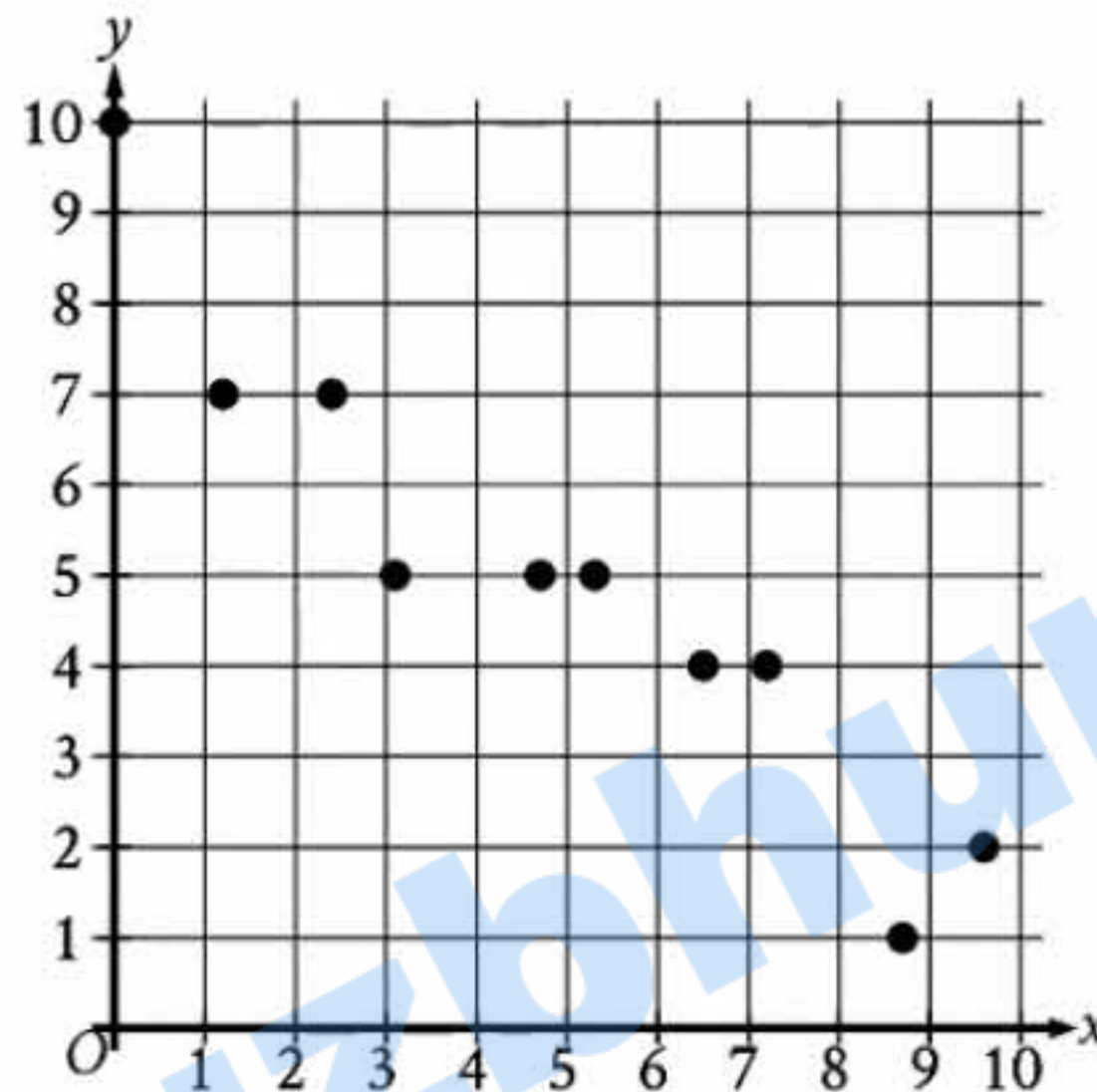


Reference



More

The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = 0.8 + 8.7x$

(B) $y = 0.8 - 8.7x$

(C) $y = 8.7 + 0.8x$

(D) $y = 8.7 - 0.8x$



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Calculator



Reference



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8



Mark for Review



The function $f(x) = \frac{1}{9}(x - 4)^2 + 5$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- (A) The toy car's minimum height was 5 inches above the ground.
- (B) The toy car's minimum height was 4 inches above the ground.
- (C) The toy car's height was 5 inches above the ground when it started moving.
- (D) The toy car's height was 4 inches above the ground when it started moving.



Hide

9

Mark for Review

$$7x - 5y = -70$$

For the given equation, which table gives three values of x and their corresponding values of y ?

- (A)

x	0	5	10
y	14	21	28
- (B)

x	0	5	10
y	28	21	14
- (C)

x	14	21	28
y	0	5	10
- (D)

x	14	21	28
y	10	5	0



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10

Mark for Review

A chemist combines water and isopropanol to make a mixture with a volume of 45 milliliters (mL). The volume of isopropanol in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)

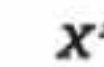
- A) 55 B) 35 C) 45 D) 10



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Calculator



Reference



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11



Mark for Review



$$y = -\frac{1}{7}x$$

$$y = \frac{1}{11}x$$

The solution to the given system of equations is (x, y) . What is the value of x ?

(A) -7

(B) 0

(C) 4

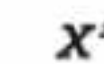
(D) 11



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12



Mark for Review



A company has a customer loyalty program. In January 2018, there were 1,600 customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 5% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c , enrolled in the company's loyalty program m months after January 2018, where $m \leq 24$?

(A) $c = 1,600(0.05)^m$

(B) $c = 1,600(1.05)^m$

(C) $c = 1,600(1.5)^m$

(D) $c = 1,600(5)^m$



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13

Mark for Review

The function $f(t) = 10,000(2)^{\frac{t}{850}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A) 0 B) 850 C) 350 D) 425

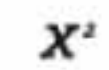
Answer Preview:



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Calculator



Reference



More

14



Mark for Review



$$y = \frac{5}{7}x + 9$$

One of two equations in a system of linear equations is given. The system has infinitely many solutions. What is the slope of the graph of the second equation?

(A) $-\frac{7}{5}$

(B) $-\frac{5}{7}$

(C) $\frac{5}{7}$

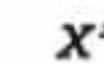
(D) $\frac{7}{5}$



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Calculator



Reference



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15



Mark for Review



If $4x^2 - 12x - 24 = 0$, what is the value of $x^2 - 3x$?

(A) 0

(B) 3

(C) 4

(D) 6

16.

The function f is defined by $f(x) = 41(0.28)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?

Ⓐ 72

Ⓑ 59

Ⓒ 41

Ⓓ 28



Hide

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17

Mark for Review

Raheem bought 9 shirts that were each the same price. He used a coupon for \$54 off the entire purchase. The cost for the entire purchase after using the coupon was \$108. What was the original price, in dollars, for 1 shirt?

- A) 12 B) 15 C) 18 D) 21

18.

To cut a lawn, Antwan charges a fee of \$15.00 for his equipment and \$8.50 per hour spent cutting a lawn. Taylor charges a fee of \$12.00 for her equipment and \$9.00 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are all the values of x for which Taylor's total charge is greater than Antwan's total charge?

Ⓐ $5 \leq x \leq 6$

Ⓑ $6 \leq x \leq 7$

Ⓒ $x < 5$

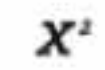
Ⓓ $x > 6$



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Calculator



Reference



More

19



Mark for Review



In right triangle ABC , angles A and B are acute, side AC has a length of 21.2, and $\tan B = \frac{1}{6}$. What is the length of side BC , rounded to the nearest tenth?

(A) 3.5

(B) 4.6

(C) 127.2

(D) 449.4

x	$f(x)$
-9	155
-3	227
3	155

Three points on the quadratic function $f(x)$ are given in the table. If $g(x) = f(x + 4)$, what is the y -intercept of the function $g(x)$?

- A) 132 B) 129 C) 126 D) 127



Hide



Calculator



Reference



More

21



Mark for Review



$$f(x) = (x - 4)^2 + 11$$

What is the minimum value of the given function?

(A) 4

(B) 7

(C) 11

(D) 15



Hide



Calculator



Reference

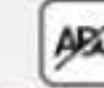


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22



Mark for Review



Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

(A) $\frac{25\pi}{2}$

(B) $\frac{27\pi}{2}$

(C) 22π

(D) 23π



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1

Mark for Review

$f(x) = 8x + b$

For the linear function f , b is a constant and $f(0) = 11$. What is the value of b ?

- A) 0 B) 11 C) 13 D) -11

Answer Preview:



Hide



Calculator



Reference



More

2



Mark for Review



Data set A: 5, 8, 11, 11, 14, 20

Data set B: 5, 8, 11, 11, 14, 20, 41

Two data sets are shown. Which statement best compares the medians of the data sets?

- ☐ (A) The median of data set A is greater than the median of data set B.
- ☐ (B) The median of data set A is less than the median of data set B.
- ☐ (C) The medians of data sets A and B are equal.
- ☐ (D) There is not enough information to compare the medians.



Hide



Calculator



Reference



More

3



Mark for Review



To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.2, with an associated margin of error of 0.08. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

- (A) It is plausible that the proportion is between 0.12 and 0.28.
- (B) It is plausible that the proportion is less than 0.12.
- (C) The proportion is exactly 0.2.
- (D) It is plausible that the proportion is greater than 0.28.



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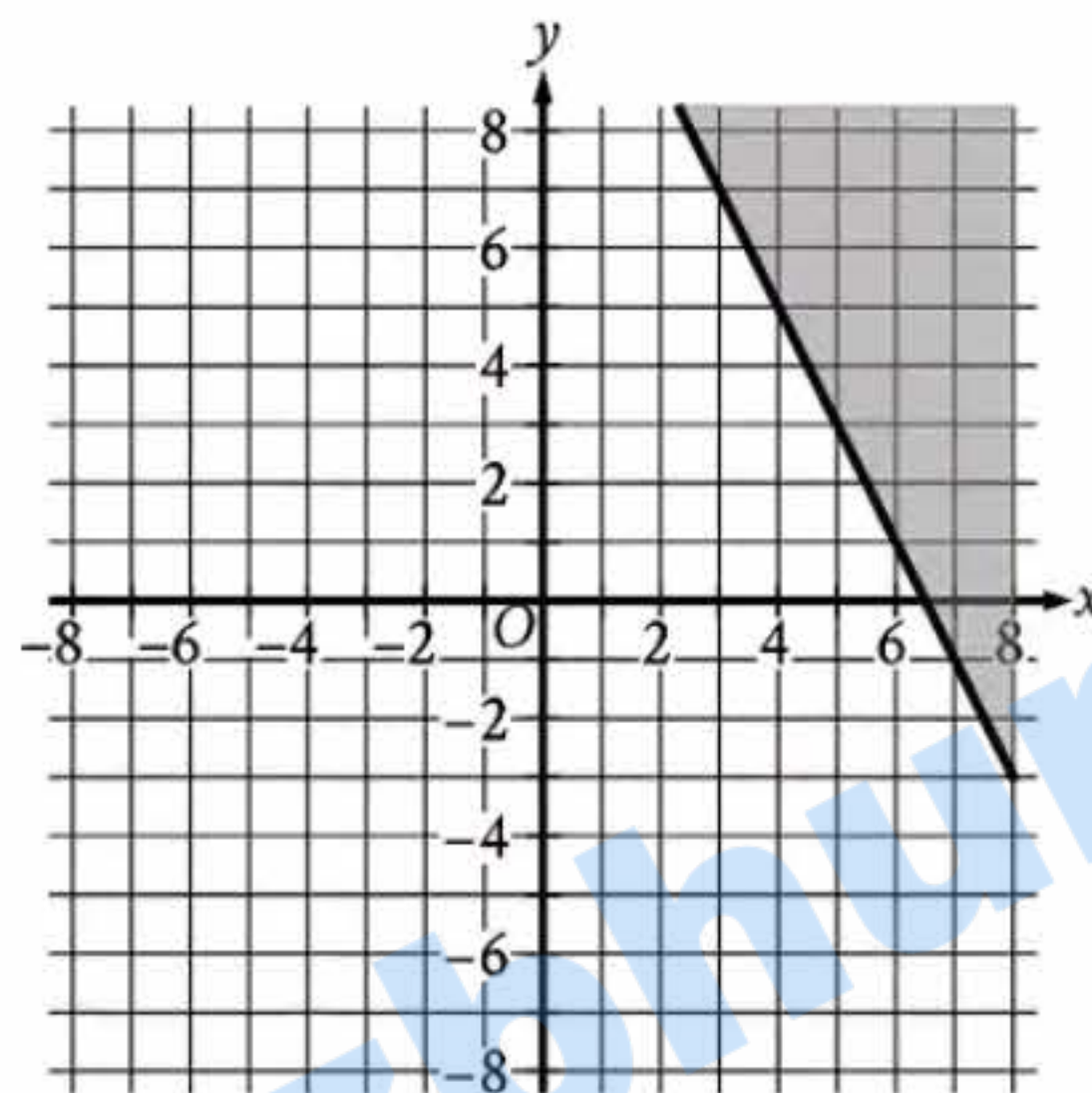
Calculator



Reference



More



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

(A) $(-8, 0)$

(B) $(0, 8)$

(C) $(0, -8)$

(D) $(8, 0)$



Hide



Calculator



Reference



More

5



Mark for Review



x	$f(x)$
-1	4
0	12
1	22

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

(A) $f(x) = 5x^2 + 5x + 12$

(B) $f(x) = 9x^2 + x + 12$

(C) $f(x) = 9x^2 - x + 12$

(D) $f(x) = x^2 + 9x + 12$



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Calculator



Reference

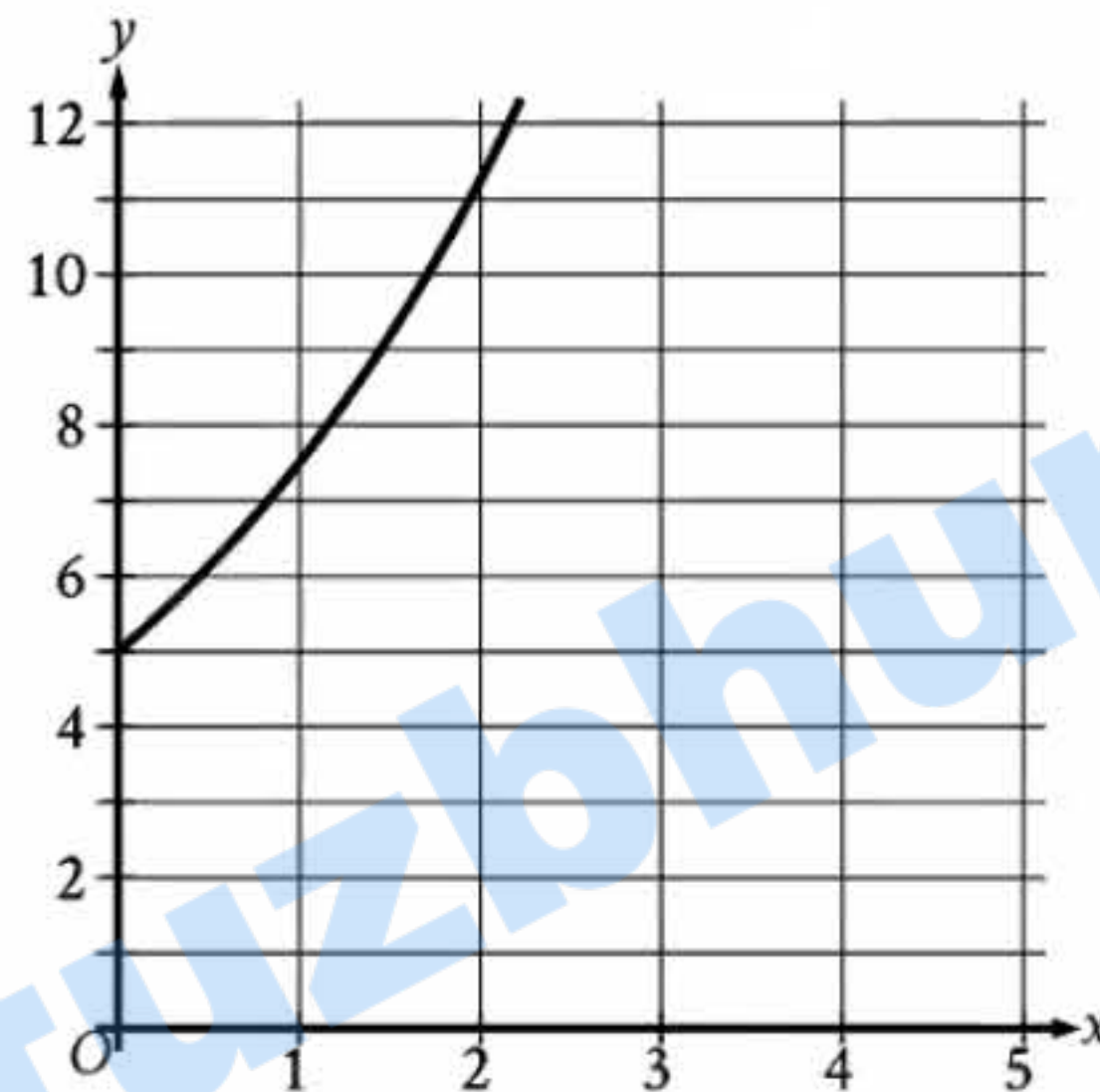


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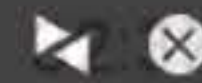


Mark for Review



The graph gives the estimated population y , in thousands, of a town x years since 2007, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

- (A) The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.
- (B) The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.
- (C) The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.



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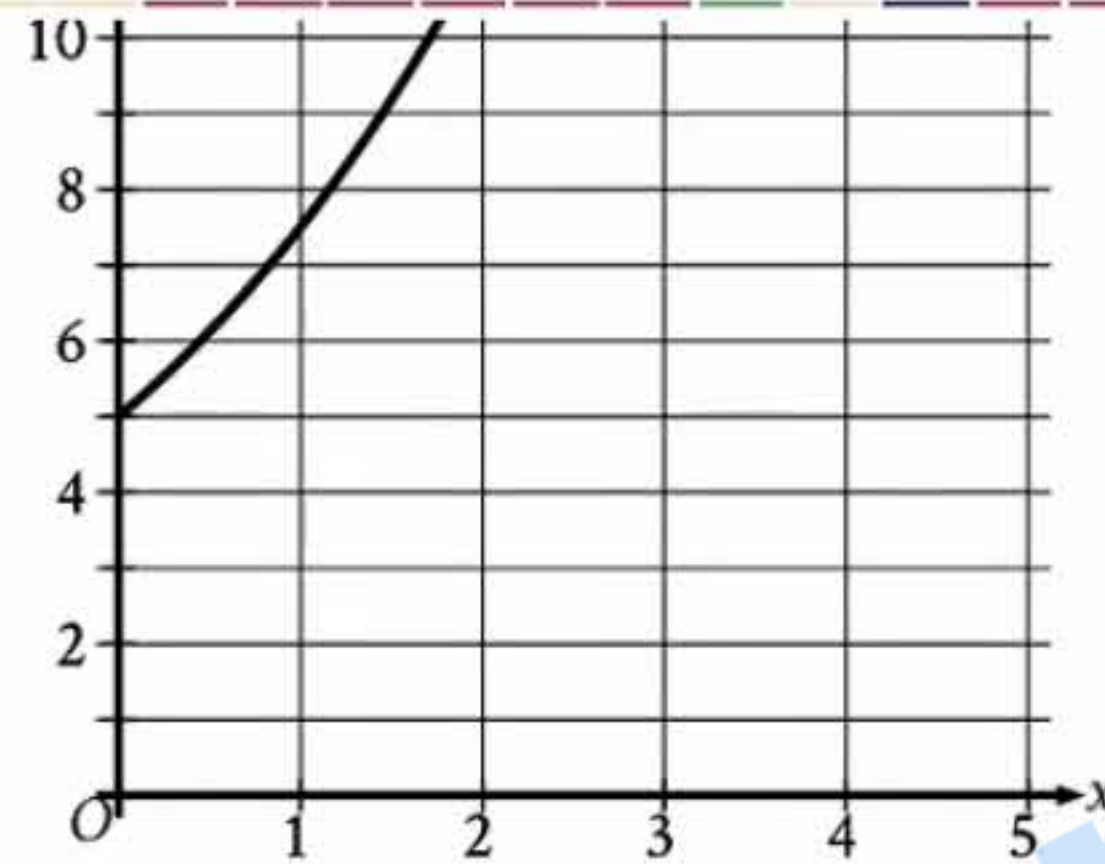
Calculator



Reference



More



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Calculator



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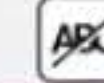


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7



Mark for Review



$$T(x) = 62 - 8x$$

The function T estimates the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), in a certain city one day in January, where x is the number of hours after 10 a.m. and $1 \leq x \leq 5$. According to the function, what was the estimated decrease in temperature each hour, in $^{\circ}\text{F}$, in this city during this time period?

(A) 4

(B) 8

(C) 54

(D) 62



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Calculator



Reference

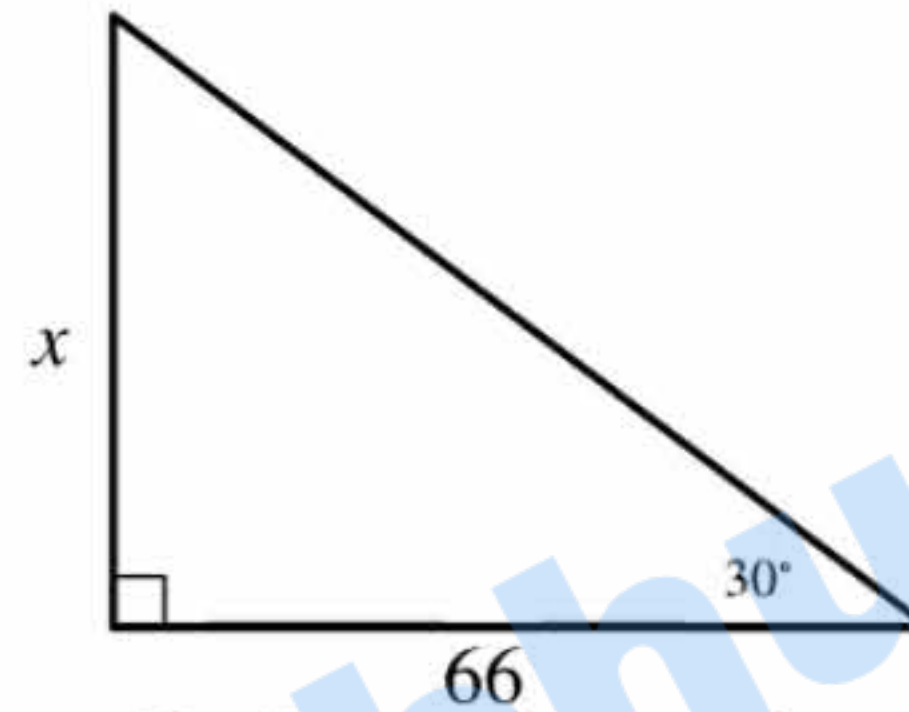


More

8



Mark for Review



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

(A) 22

(B) $22\sqrt{3}$ (C) $66\sqrt{3}$

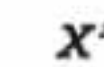
(D) 132



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Calculator



Reference



More

9



Mark for Review



There is a proportional relationship between p and t . If $p = 2,340$ when $t = 3,120$, then what is the value of p when $t = 2,496$?

(A) 1,716

(B) 1,872

(C) 3,276

(D) 3,328



Hide

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10

Mark for Review

A cube has a volume of 79,507 cubic units. What is the surface area, in square units, of the cube?

A) 1849 B) 7396 C) 11094 D) 5547

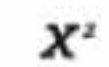
Answer Preview:



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Calculator



Reference



More

11



Mark for Review



$$\frac{8(cx+3)}{5} = 4x - 3$$

In the given equation, c is a constant. If the equation has no solution, what is the value of c ?

(A) $-\frac{5}{2}$

(B) $-\frac{2}{5}$

(C) $\frac{2}{5}$

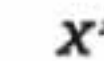
(D) $\frac{5}{2}$



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Calculator



Reference



More

12



Mark for Review



The function $f(x)$ is defined as 17 more than 3 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

- (A) When $f(x) = 0$, the number is $-\frac{17}{3}$.
- (B) When the number is 0, $f(x) = 17$.
- (C) For each increase of 1 in the value of the number, $f(x)$ increases by 3.
- (D) The value of $f(x)$ increases by 1 for each increase of 3 in the value of the number.



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Calculator



Reference



More

13



Mark for Review



$$\frac{2}{c} - \frac{7}{d} = 5$$

The given equation relates the variables c and d , where c and d are greater than 1. Which equation correctly expresses d in terms of c ?

(A) $d = \frac{c}{7(2-5c)}$

(B) $d = \frac{7c}{2-5c}$

(C) $d = \frac{7(2-5c)}{c}$

(D) $d = \frac{2-5c}{7c}$



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- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 ~ ~ ~ ~	0.66 .66 0.67

14

Mark for Review

A proposal for a construction project was included on a city election ballot. A TV news report stated that 8 times as many people voted in favor of the proposal as people who voted against it. An internet article reported that 24,500 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

- A) 21000 B) 3500 C) 16500 D) 8000

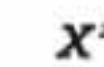
Answer Preview:



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Calculator



Reference



More

15



Mark for Review



The function f is defined by $f(x) = 4x^2$. What is the value of $f(11)$?

(A) 44

(B) 52

(C) 88

(D) 484



Hide

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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16

Mark for Review

$$\sqrt[5]{p^2} = t^{\frac{3}{4}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{2n-1}$, where n is a constant, what is the value of n ?

- A) $13/5$ B) $23/30$ C) $3/5$ D) $3/10$

Answer Preview:





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17



Mark for Review



The equation $4|x - 3| = k$, where k is a constant, has exactly one solution. Which of the following could be the value of $\frac{k}{4}$?

(A) -3 only

(B) -3 or 3

(C) 0 only

(D) 3 only



Hide

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18

Mark for Review

$$-4x^2 + bx - 25 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

- A) 20 B) 19 C) 21 D) -19



Hide



Calculator



Reference



More

19



Mark for Review



The function f is defined by $f(x) = a(3.1^x + 3.1^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane as a constant or coefficient?

I. $g(x) = a(3.1^x + k)$

II. $h(x) = a(3.1)^x + m$

☐ (A) I only

☐ (B) II only

☐ (C) I and II

☒ (D) Neither I nor II



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Calculator



Reference



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Mark for Review



The cost of renting a piece of construction equipment for up to 5 days is \$230 for the first day and \$115 for each additional day. Which of the following equations gives the cost y , in dollars, of renting the equipment for x days, where x is a positive integer and $x \leq 5$?

(A) $y = 230x + 115$

(B) $y = 230x - 115$

(C) $y = 115x + 230$

(D) $y = 115x + 115$



Hide

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21

Mark for Review

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

- A) 4980 B) 60000 C) 49800 D) 41334

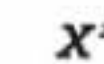
Answer Preview:



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Reference



More

22



Mark for Review



For triangle PQR and triangle STU , angles P and S each measure 30° , $QR = 16$ and $TU = 12$. Which additional piece(s) of information is(are) sufficient to prove that triangle PQR is similar to triangle STU ?

- I. The length of $\overline{PQ}$ is $\frac{4}{3}$ the length of $\overline{ST}$.
II. The length of $\overline{PR}$ is $\frac{4}{3}$ the length of $\overline{SU}$.

- (A) I is sufficient, but II is not.
- (B) II is sufficient, but I is not.
- (C) I is sufficient, and II is sufficient.
- (D) Neither I nor II is sufficient.